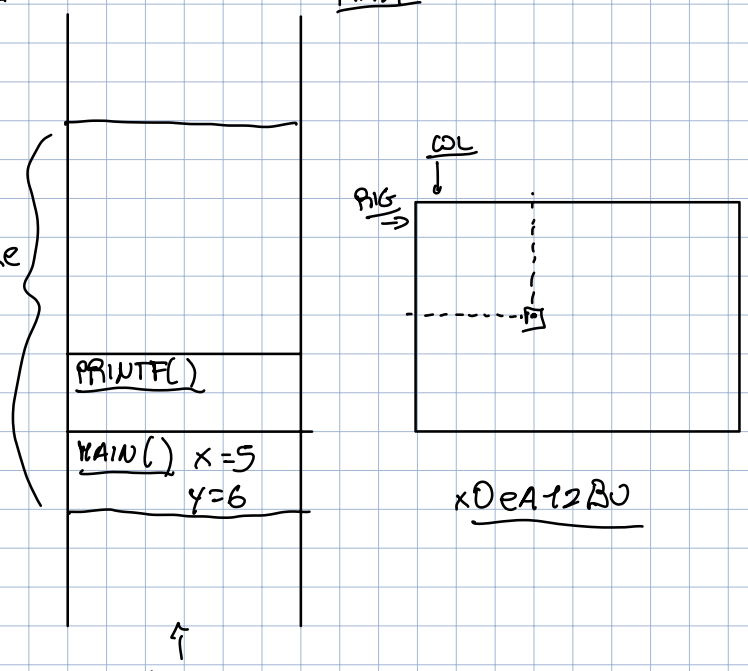
esl.c

```
int main(){
  int x=5;
  int y=6;
  printf(...);
```

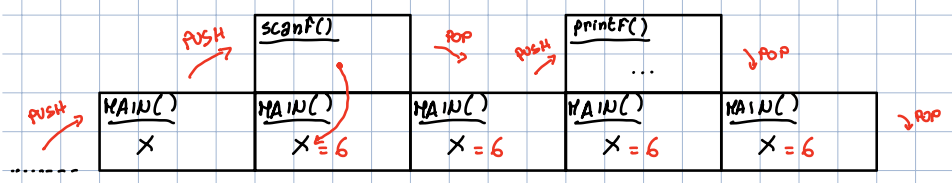


STACK è LIFO

(LAST IN - FIRST OUT)

```
#include ....
int main(){
  int x;
  scanf("%d", &x);
  printf("%d\n", x);
}
```

EVOLUZIONE DELLO STACK



ES 1 PIPELINE

→ n° ISTRUZIONI = 5

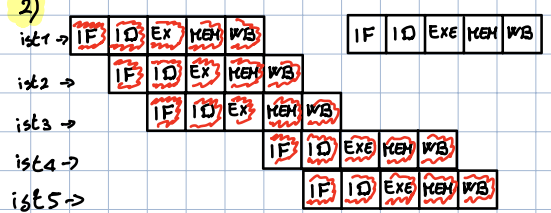
→ n° PIPELINE = 3

→ T = 0,5 ns

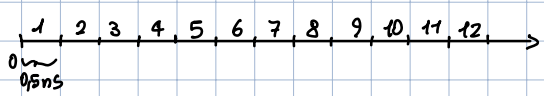
1) TEMPO SENZA PIPELINE?
2) TEMPO CON PIPELINE?
3) DOPO QUANTO TEMPO INIZIA LA LA SECONDA ISTRUZIONE?
4) TEMPO RISPARMIATO TRA I DUE METODI?
5) MIPS CON PIPELINE?

1) $\underset{\substack{\uparrow \\ \text{n° ISTRUZIONI}}}{5} \cdot \overset{\substack{\text{FASI DEL CICLO MACCHINA} \\ \text{X SINGOLA ISTRUZIONE} \\ \downarrow}}{5} \cdot 0{,}5\,ns = 12{,}5\,ns$

2)



TEMPO CON 3 PIPELINE: 5,5 ns



3) 0,5 ns = TEMPO DI INIZIO PER LA SECONDA ist

4) $T_{SENZA\,P} = 5 \cdot 5 \cdot 0{,}5 = 12{,}5\,ns$

$T_{CON\,P} = 5{,}5\,ns = (11 \cdot 0{,}5\,ns)$

$T_{RISPARMIO} = 12{,}5 - 5{,}5 = 7\,ns$

5) $MIPS_3 = \dfrac{5}{5{,}5 \cdot 10^{-9}} =$

↑ second

$5 : 5{,}5\,ns = x : 1\,s$

$x = \dfrac{5 \cdot 1\,s}{5{,}5\,ns} = \dfrac{5 \cdot 1\,s}{5{,}5 \cdot 10^{-9}\,s} =$

$MIPS_1 = \dfrac{5}{12{,}5 \cdot 10^{-9}} = \dfrac{5 \cdot 1\,s}{12{,}5 \cdot 10^{-9}\,s} =$

VANTAGGIO DELLO SCORE = $MIPS_3 - MIPS_1$